

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

B.Tech 3rd Semester — Course Assignments

Subject: Operating System

Subject Code: B109313(022)

Assignment 01: Unit 01 — Introduction to Operating System

1. Explain structure of operating system.
2. What are the services of operating system?
3. Explain Time sharing and real time operating system?
4. Describe Evolution of OS?
5. Describe Distributed Computing and Parallel computing with example.

Assignment 02: Unit 02 — Process Management

1. Describe the objectives of scheduling. Explain structure of PCB.
2. What are process states? Describe in brief.
3. Write short notes on:
 - Producer Consumer Problem
 - Reader Writer Problem
 - Dining Philosopher Problemwith suitable diagrams.
4. What is Critical section problem in detail?
5. For the following table draw a chart illustrating their execution using:
 - (i) First Come First Serve (FCFS)
 - (ii) Shortest Remaining Time Next (SRTN)
 - (iii) Round Robin (quantum time = 2 ms)

Process	Arrival Time	Processing Time
P0	0.0	3
P1	1.0	6
P2	4.0	4
P3	6.0	2